Measurement Kick as a Scattering Event

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We develop a minimal model of a single measurement event between a qubit and a detector particle...

I. INTRODUCTION

Santiago comments/edits , Jeff comments/edits , Oded comments/edits

- Saving intro for the end.

The measurement problem is one of the most important unanswered questions in quantum mechanics. At the heart of the matter, is the collapse postulate [1], which introduces non-unitary stochastic jumps to the otherwise deterministic, unitary evolution predicted by the Schrödinger equation.

Already in 1932, John Von Neumann proposed an extension to the Copenhagen interpretation [2], by adding the detector and its interaction with the target system to the Hamiltonian as quantum objects.

Although this does not solve the underlying problem, it has resulted in novel effects such as non-projective, continuous measurements [3], and weak value experiments [4, 5], providing a more general theory of measurement beyond collapse of the wave function.

Quantum Point Contacts are charge detectors, ~~exhibiting quantized conductance~~ [6–9] capable of realizing such non-invasive measurements [10]. Explain the working principle.

Experimental realizations and advantages of Quantum Point Contact. These studies evidence that QPCs have the potential to carry out both information processing and control protocols in a quantum computing setup.

It has been shown that, the current flowing through QPCs becomes correlated to the state of the qubit [11–13].

Using the rate equation formalism, it has been shown that QPC measurements can achieve fine control over quantum dot systems [14, 15] and are adequate detectors of coherent transport [16, 17].

What is missing from the rate equations that may be captured in single events.

Existing studies ... (Here we should look at Klaus Mølmer's work)

Therefore, it is desirable to study the effects of a QPC measurement on a qubit for a single event.

In this work, we develop a minimal model of a single measurement event between a qubit and a detector particle. Using exact diagonalization, together with a perturbative analysis and the ancilla paradigm, we find an optimal regime for measurement and identify the mechanisms behind entanglement generation.

II. HAMILTONIAN AND MEASUREMENT SETUP

- Introduce the "unraveled Hamiltonian" in Eq.(1), sketched in Figs.1a and 1b, and define all quantities.
- The decoupled energies are Eq.(2).
- Composite initial state is a product state Eq.(4).
- Initial condition for QPC is a wavepacket (5).
- Initial state for qubit ensures orbit is within the x-z plane.
- We keep our treatment purely Hamiltonian and unitary, so observables like entropy are readily available.

$$\hat{H} = -J\sum_{j=1}^{N}\hat{c}_{j+1}^{\dagger}\hat{c}_j - \omega\hat{d}_1\hat{d}_0 + g\hat{d}_1^{\dagger}\hat{d}_1\hat{c}_{b+1}^{\dagger}\hat{c}_b + \text{h.c.} \tag{1}$$

$$E(k) = -J\cos k0 \tag{2}$$

$$\epsilon_{\pm} = \pm\omega \tag{3}$$

$$|\Psi\rangle = |\psi\rangle \otimes (\alpha_0|0\rangle + \beta_0|1\rangle) \tag{4}$$

$$|\psi\rangle = \left(\frac{1}{\pi\Delta^2}\right)^{1/4} e^{-\frac{1}{2\Delta^2}(x_j-x_c)^2+ik_0(x_j-x_c)}\left|\vec{j}\right\rangle \tag{5}$$

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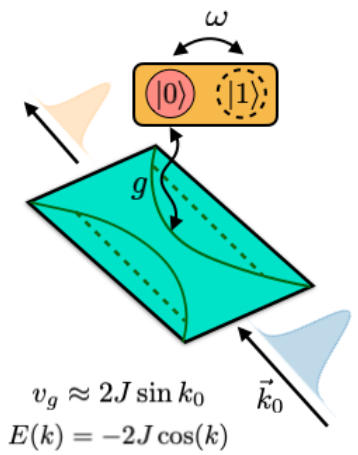
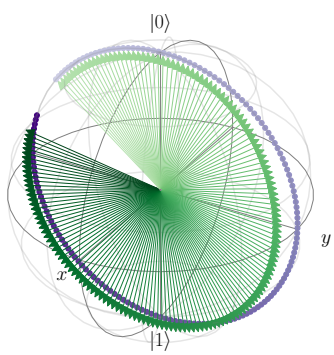


(a)

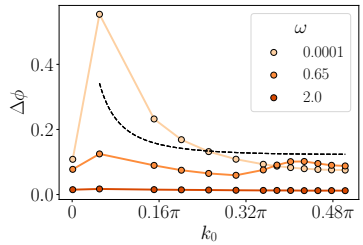


(b)

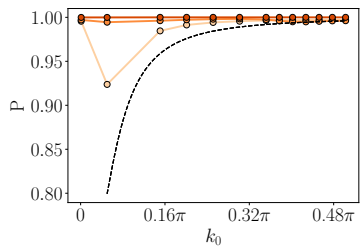
FIG. 1. (a) Diagram of the model in Eq.(1), the QPC particle is represented as a Gaussian wavepacket and the qubit as a 2-level system. (b) Bloch sphere trajectories for the qubit I should include the theta and phi angles in the figure to also show the notation . The free evolution corresponds to the purple dots and the QPC influenced one to the green arrows

III. COMPARISON TO OTHER MODELS

- Wavepacket allows for the notion of a measurement time ($\delta\tau = \Delta(k_0)/v_g$) in terms of its spread which is dependent on k_0.
- Comparing k_0 to the Rabi frequency allows us to define different regimes.
- In the limit of a slow (quasi-static) qubit (with respect to k_0), measurement according to von Neumann.
- VN is a unitary Hamiltonian approach for a single measurement event, where the detector is a free wave-packet and the the qubit's dynamics are deprecated during the interaction window.
- VN produces an entangled state, where the momentum of the detector is coupled to the system's measured observable.
- Hence, measurement is done by reading out the detector state with a projection, which yields partial information on the entangled observable. Through this, VN provides a platform for weak-measurements.
- An extension of VN to multiple measurement kicks in a QPC is found in K [18] and G [12].
- In K and G, it is assumed that, i) the qubit is slow enough to be considered constant over a single kick, ii) the detector and system are weakly coupled. These are the same assumptions as VN.
- These assumptions alow a formulation where the readout observable is the current through the QPC.
- ~~K and G obtain Bloch-like (classical) equations of motion for the qubit density matrix in terms of the QPC transmission rate.~~
- The current becomes dependent on qbit's occupation and agrees with Landauer (Fermi's golden rule zeroth order effect). do we include the equaion here?
- The backaction on the qubit is manifested as dephasing (exponential decay of coherences) and a slow down of the Rabi frequency.
- in the quasi-static limit, our results should be consistent with VN and KG, we also expect such behavior in our model.
- In addition, we expect a phase gain since the interaction and free Hamiltonian's do not commute (we break non-demolition) [19].
- In the opposite limit of fast qubit (Floquet), we expect decoupled dynamics as the qubit's oscillations are too fast to be affected by a single kick and the QPC wavepacket will experience the qubit as a constant potential.
- However, in the intermediate (coherent) regime, the oscillations of the QPC will be important for a single kick.
- Here, one can expect that k_0/ω will play a role in the measurement outcome, but its difficult to get a priori intuition as its is not a commonly discussed.
- We can explore all of these regimes by tuning the Rabi frequency and v_g in our model Eq.(1).
- We can solve the exact dynamics with ED (details in A).
- Notice that, our model is equivalent to measurement without readout so there is no projection applied on the QPC.
- ED allows us to get the exact backaction from the reduced density matrix on the qubit, which we can contrast to von Neumann/KG.



(a)



(b)

FIG. 2. Backaction on the qubit as a function of the central momentum k_0 for different Rabi frequencies ω. The orange dots are data from exact diagonalization and the dashed line is the von Neuman analytical prediction from Eq.(7). (a) Change in the qubit's phase due to measurement (b) Qubit's purity after measurement.

IV. RESULTS

A. Measurement Backaction

- With ED, we obtain the exact qubit density matrix, and the Bloch angles θ, ϕ at all times. Details in A.
- The backaction after measurement, is shown in Fig.2 with the phase gain $\Delta\phi$ Fig.2a and purity Fig.2b as a function of k_0. Should we mention something about the damping of qubit oscillations?
- Data with low k_0 enhances the backaction, but the kick becomes ill-defined below $k_0 \approx 0.16\pi$.
- Three distinct behaviors in ω: Slow qubit $\omega = 0.0001$ which agrees with VN and dephasing, fast qubit ($\omega = 2.0$) in which QPC and qubit are decoupled (no loss of purity or change in phase), and intermediate ($\omega = 0.65$) where backaction is non-monotonic.
- We do not see a change in Rabi frequency (See figure in Appendix A) as this is a higher order effect.
- These ED results can be understood with a toy model employing VN's approach [20] in the continuum limit.
- Before measurement, the dynamics are described by the free Hamiltonians of each independent system [19].
- At measurement, the time evolution operator is approximated by Eq.(6) for small ω [2].
- The resulting effective measurement strength $\tilde{g} = g\delta\tau$ depends on the time window $\delta\tau$.
- Applying Eq.(6) results in an entangled state, coupling the $|1\rangle$ qubit state to the position of the wave packet (See B). These are the so-called meter states [21].
- Maybe I should include the entangled state equation here .
- Tracing out the detector yields the qubit's reduced density matrix Eq.(7) in terms of the new Bloch vector Eq.(8), valid to second order in g [19].
- $\rho^{'}$ shows that the Rabi frequency remains unchanged and the qubit gains a phase $\Delta\phi(k_0) = -gk_0\delta\tau(k_0)$, where $\delta\tau$ is proportional to the spread of the wave-packet at the time of interaction (mentioned above).
- (discussion of the units may be necessary here)
- As mentioned, the change in phase is expected, since we break non-demolition.
- Purity is lost as $P = 1/2(1+R^2\sin^2(\theta_0)+\cos^2(\theta_0))$, with $R = e^{-g\delta\tau/4\Delta^2}$, consistent with dephasing measurement.
- The equations for $\Delta\phi$ and k_0 show that k_0 tunes the effective interaction strength by controlling the interaction time, in turn modifying $\tilde{g}$.
- Our results imply that, $k_0 \to \pi/2$ to remain within the weak measurement regime .
- VN fails to predict results for $\omega = 0.65$ and $\omega = 2.0$, however it evidences the role that k_0 plays in defining the effective interaction strength.
- In the following, we situate the three observed behaviors in distinct entropy regimes, with a perturbative expansion.

$$\hat{U} = e^{-ig\delta\tau\hat{n}_1\otimes\hat{p}}, \tag{6}$$

$$\rho'_q = \frac{1}{2}(\mathbb{1} + \vec{q}^{'}\cdot\vec{\sigma}) \tag{7}$$

$$\vec{q}^{'} = \begin{pmatrix} R\sin(\theta_0)\cos(\phi+\Delta\phi(k_0)) \\ R\sin(\theta_0)\sin(\phi+\Delta\phi(k_0)) \\ \cos(\theta_0) \end{pmatrix} \tag{8}$$

B. Entanglement Entropy

- The ED results in Fig.3a shows the post-measurement entropy as a function of both k_0 and ω. It is normalized by the singlet entropy.
- Highlight that entropy is an observable not discussed in the rate equation K,G formalism.
- Three main regions: I. Scattering regime: Entanglement constant in ω. II. Coherent regime: entanglement exhibits a resonant line with enhanced entanglement. III. Floquet regime: Zero entropy, system and detector are decoupled.
- Below the horizontal dashed green line, there is no weak measurement and the kick is ill defined.
- Appart from the final value of the entropy, each regime also has distinct time evolution (Fig.3b).
- Importantly, in region II (coherent), we see oscillations whenever the qubit crosses the x-y plane on the bloch sphere
- Despite these differences, the QPC transmission exhibits only a slight change in its final value, indicating that the net result is an enhanced measurement strength.

$$\left|\psi_\nu^{(1)}(k)\right\rangle = \frac{1}{N+1}\sum_{p\neq k}|p\rangle\otimes\left\{\frac{\xi(k,p)}{-2J(\cos k-\cos p)}|\nu\rangle + \frac{\xi(k,p)}{2\epsilon_\nu - 2J(\cos k-\cos p)}|\mu\rangle\right\} + \frac{1}{N+1}\frac{\xi(k,k)}{2\epsilon_\nu}|k\rangle\otimes|\mu\rangle \quad (9)$$

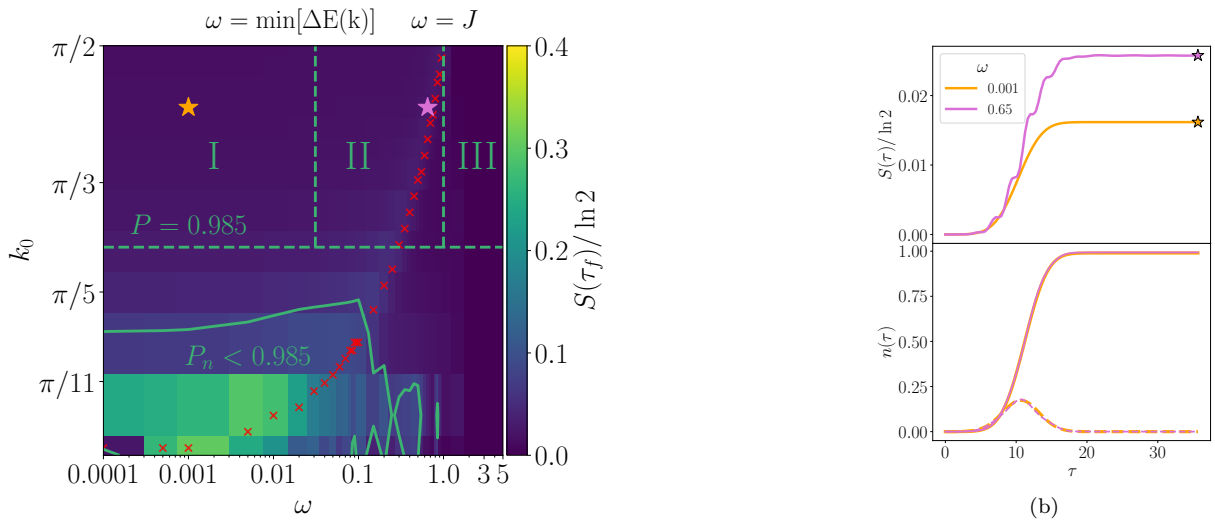
$$\omega < \min\Delta E(k) \quad (10a)$$

$$\omega > J \quad (10b)$$

- The processes that dominate each of these regions are identified with each term in Eq.(9). Derived in Appendix C.
- Specifically, we are are looking for processes that mix different bands Why can we do this? I think we can justify our interpretation in terms of scattering from the point of view of the S-matrix which should correspond to these three processes too .
- The first term is similar to a tight binding model with constant external potential and is degenerate only towards the edges of the Brillouin zone (we are not interested in these). Mostly relevant to regime III in 3a.
- Since it represents events where scattering happens only between states inside the same band, we have no entanglement.
- The third term mixes between band gaps, but is only degenerate when the Rabi frequency approaches zero, therefore it only becomes relevant in regime I.
- Here, we have a scattering process between two bands without momentum exchange. Entropy production is influenced purely by the size of the time window.
- Finally, we have the second term which also mixes between two different bands ($\mu \neq \nu$). It is degenerate whenever the difference between two momenta is equal to the band gap.
- This is only possible inside regime II, as dictated by the bounds Eq.(10) plotted in Fig.3a.
- The second term also shows that the resonant line corresponds to $2\epsilon_\nu = \Delta E(k)$ where the symmetric and anti-symmetric states are strongly hybridized.
- The resonant line is a feature of the eigenenergies, but in our setup its only becomes relevant when k_0 crosses it. This explains the non-monotonic behavior in Fig.2.
- One can thing of this in analogous terms to hydronic excitations. The difference is that, here the energy transfer goes to/from the qubit
- This sort of transfer of energy through measurement has been observed in cotunneling [15] .
- We note that, the resonant line is not a finite size effect, but generic in large systems. In fact it is more pronounced in long chains: as momentum states become more dense, we will always find two whose energy difference is the same as the band gap.

V. DISCUSSION AND OUTLOOK

- We propose a simple Hamiltonian model for the QPC-qubit measurement process, that allows us to go beyond the typical treatment.
- Our model allows a well defined weak-measurement kick in terms of the width of a Gaussian wave packet and the post-measurement purity.
- We show that the main parameters are k_0 and ω, and ideintify their range.



(a) (b)

FIG. 3. (a) Post-entanglement Entropy (Normalized by the singlet-entropy) as a function of wave-packet momentum and Rabi frequency. The dashed, green vertical lines bound the regimes according to Eq.(10) and the dashed horizontal line marks where the purity from Eq.(7) falls below 0.985. The solid line indicates where the purity from the exact diagonalization P_n falls bellow that same threshold. In addition, the red dots show the resonant line, where the qubit energy becomes degenerate with the QPC's kinetic energy. (b) add one for regime III Top: Time evolution of the entanglement entropy. Bottom: Time evolution of the QPC transmission to the right of the bond (solid line) and the occupation directly at the bond (dashed line). The purple and orange lines are color coded to correspond to the points marked by stars in (a).

- From the entanglement entropy and backaction on the qubit, we found 3 regimes: Static, coherent and Floquet, and identified the entropy generating processes via the degeneracies of the perturbative corrections.

- In the quasi-static regime, we reproduced the main features of non-invasive QPC measurements from VN, K and G: Our transition agreed with VN/Landauer and purity loss is consistent with dephasing in continuous measurements. Rabi change?

- In this regime, entropy is generated when the symmetric and anti-symmetric states become degenerate.

- On the other hand, we show that in Floquet regime, the QPC and qubit become decoupled.

- This is reflected both in the backaction and the entropy. The later is explained by perturbative terms similar to a tight-binding chain interacting with an external, constant potential, which is only relevant for slow wavepackets (edges of Brillouin zone).

- Finally, the coherent regime exhibits unique behavior, characterized by a resonant line resulting in enhanced measurement strength.

- This resonant line indicates that more energy is absorbed/emitted by the system through a new scattering channel not present in typical calculations.
- Now follow up with Outlook ideas

- Rabi frequency is constant so this can be a way of implementing robust phase gates

- The phase change is also seen in [22] and we could also extend our model to several subsequent measurements which could be used to prove Legget-Garg inequalities.

- Another extension is including spin and valley degrees of freedom, relevant to sylicon based quantum computing .

VI. RDM STATEMENT

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Appendix A: Numerical Setup and Finite Size Effects

- For our numerical simulations it is convenient to express the Hamiltonian Eq.(1) in terms of the computational basis Eq.(A1).
- The term $\left|\vec{j}\right\rangle$ indicates a particle in the j-th site of the QPC chain, $|q\rangle = |0\rangle, |1\rangle$ represents the qubit state as a double dot.
- As the composite state includes a Gaussian wavepacket in the QPC, the initial spread Δ has to be chosen such that the spread in momentum space $\Delta_p = \hbar/(\sqrt{2}\Delta)$ is strongly peaked around the central momentum k_0, for clearly defined trajectories [23].
- This allows us to calculate the traveling time of the QPC particle with Ehrenfest's theorem $\tau = \langle \delta x \rangle / v_g$ where δx is the distance (in terms of lattice spacing) and $v_g = 2J \sin k_0$ is the group velocity.
- With this relation we can choose the initial state (in Eq.(4)) of the qubit, so it is always in the same occupation (parametrized by θ_f) when the QPC hits the interaction site, regardless of v_g Eq.(A2).
- This condition also ensures that the qubit always traces the same Bloch sphere trajectory, oscillating between both poles.
- In addition, Ehrenfest's theorem helps in avoiding reflection effects, stopping the simulations when the wavepacket hits the end of the chain.

$$\left|\vec{j}\right\rangle \otimes |q\rangle = |00\ldots010\ldots0\rangle \otimes |q\rangle \tag{A1}$$

$$\alpha_0 = \cos(\theta_0/2) \tag{A2}$$
$$\beta_0 = -i\sin(\theta_0/2) \tag{A3}$$
$$\theta_0/2 = -\theta_f/2 + \tau_B\omega \tag{A4}$$
$$\tag{A5}$$

- A standard ED time evolution algorithm was implemented in numpy, by applying the time evolution operator to the initial state in the diagonal basis to obtain the time evolved state $|\Psi(\tau)\rangle$.
- We track the occupations by applying the projectors Eq.(A6) to the composite state $|\Psi(\tau)\rangle$
- We also get the qubit's reduced density matrix $\hat{\rho}_q(\tau)$ by tracing out the detector states and obtain the purity $P(\tau) = \mathrm{Tr}\{\hat{\rho}_q(\tau)\}$ and von Neumann entropy $S(\tau) = -\mathrm{Tr}\{\hat{\rho}_q(\tau)\log\hat{\rho}_q(\tau)\}$.
- From $\hat{\rho}_q(\tau)$ we can obtain the backaction, however extracting the Bloch angles from the numerical density matrix is problematic.
- To avoid issues, we estimate the change in phase from the law of cosines between a free Bloch vector $\vec{q}$ and the coupled one from ED $\vec{q}'$ Eq.(A8).
- For the final Rabi frequency we apply a fast-Fourier transform to the occupations, after the measurement.
- Include discussion about extending the qubit dynamics to get a better estimate? .
- Fig.4 shows that the change in Rabi frequency is too small to differentiate from numerical error, so we deprecate it.

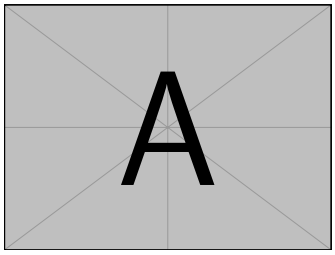

FIG. 4. (a) Change in Rabi frequency as a function of k_0 for several values of ω.

$$\hat{\Pi}_j = \left|\vec{j}\right\rangle \left\langle\vec{j}\right| \otimes \mathbb{1}_2 \tag{A6}$$

$$\hat{\Pi}_q = \mathbb{1}_N \otimes |q\rangle \langle q| \tag{A7}$$

$$\cos(\Delta\phi) = \frac{|\vec{q}|^2 + \vec{q}'|^2 - |\vec{q} - \vec{q}'|}{2\vec{q}|\vec{q}'|} \tag{A8}$$

- The parameters for the simulations are: $\Delta\tau =$, $N = 100$, $J = 1$,$\Delta = 7$, $x_c = 30$, $x_b = 50$, $\theta_f =$, $g \in \{\}$, $k_0 \in \{\}$, $\omega \in \{\}$.
- We express our results in unis of J.

Appendix B: Von Neumann's Measurement Scheme

Go into detail on the ancilla measurement scheme, and the calculation of the reduced density matrix from it Explain how von Neumann relates to ancilla measurement and talk about meter states and such. Mention the projective limit

- For the quasi-static regime, von Neumann's measurement scheme is a toy model.
- We represent the QPC particle as a free wavepacket, interacting with the qubit.
- Following [19], we separate in three intervals: Before the interaction, the interaction at $t = 0$ and after the interaction.
- We assume an initial product state
- Before, and after the interaction, system and detector follow only their respective intrinsic Hamiltonians
- During the interaction, lasting $\delta\tau$, they evolve according to the entanglement Hamiltonian as.

$$\hat{U}(\delta\tau) = e^{\int_0^{\delta\tau} d\tau H_I} \tag{B1}$$

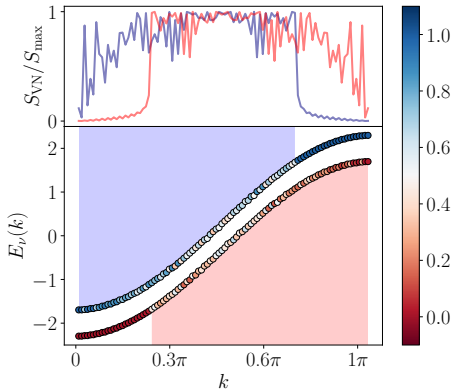

FIG. 5. Top: Entanglement entropy contributed by each eigenstate of the Hamiltonian. Bottom: Hybridization of the energy bands found via exact diagonalization. The color corresponds to the projection onto the symmetric state, the shaded regions indicate where each band becomes degenerate according to Eq. (9).

Appendix C: Derivation of the Perturbative Corrections

Derivation of the first and second order corrections to the wave function, include the eigenstates and energies of the decoupled case. Discuss why we don't get corrections to the entropy or backaction.